

Novel microstructure and properties of multicomponent CoCrCuFeNiTi_x alloys

X.F. Wang, Y. Zhang*, Y. Qiao, G.L. Chen

State Key Laboratory for Advanced Metals and Materials, University of Science and Technology Beijing,
XueYuan Road 30#, Haidian District, Beijing 100083, China

Received 10 May 2006; received in revised form 7 July 2006; accepted 4 August 2006
Available online 5 October 2006

Abstract

CoCrCuFeNiTi_x (x values in molar ratio, $x = 0, 0.5, 0.8$ and 1.0) were prepared by arc melting of the pure elements and suction casting under an argon atmosphere. Both CoCrCuFeNi and CoCrCuFeNiTi_{0.5} alloys form a single FCC solid solution. While the alloys of CoCrCuFeNiTi_{0.8} and CoCrCuFeNiTi are basically composed of primary FCC solid solution and eutectic mixture of FCC phase and Laves phase of Fe₂Ti type. The yield strength of the alloys increases from 230 MPa to 1272 MPa with the increase of Ti addition, among which the CoCrCuFeNiTi_{0.5} alloy, especially, exhibits compressive strength of up to 1650 MPa together with extensive work hardening and large plastic strain limit of 22%. An interesting transition from paramagnetism to superparamagnetism has been discovered due to the appearance of nanoparticles embedded in the amorphous phase. © 2006 Elsevier Ltd. All rights reserved.

Keywords: B. Magnetic properties; B. Mechanical properties at ambient temperature; C. Casting; D. Microstructure

1. Introduction

Conventional strategy for developing metallic alloys is to select one or two elements as the main components based on a primary property requirement, and to use alloying additions to confer secondary properties [1], such as iron-, copper-, and aluminum-based alloys [2,3] or the intermetallic compounds of Ti–Al, Ni–Al and Fe–Al binary systems which appeared in the 1970s because of their extremely high specific strengths and thermal resistance [4]. In the last two decades, multicomponent bulk amorphous alloys have attracted great attentions because they exhibit unique properties compared to its counterpart crystalline materials, e.g. 2–3 times of fracture strength of crystalline materials, high elastic strain limit and high corrosion resistance [5–9]. However, the design concept of multicomponent bulk amorphous alloys was, once again, based on one principal element [10]. Conventional alloy

development strategy leads to an enormous amount of knowledge about alloys based on one or two main components, but to very little or no knowledge about multiprincipal elements system. Actually, for quaternary, quinary and higher order systems, information about alloys in the center of the phase diagram is non-existent [1]. The main reason for not incorporating multiprincipal elements is the anticipated formation of many intermetallic compounds and complex microstructures. However, it has been reported that a quinary alloy Fe₂₀Cr₂₀Mn₂₀Ni₂₀Co₂₀ [1] and the Al_xCoCrCuFeNi alloys with minor aluminum addition [10] form a single FCC solid solution, as well with promising mechanical properties that may have potential for the requirements of high temperature alloys [11]. Just based on the simple structure and unique properties, there has been increasing interests for exploring the equiatomic multicomponent alloy series.

In this paper, a novel series of CoCrCuFeNiTi_x (x values in molar ratio, $x = 0, 0.5, 0.8$ and 1.0) alloys were prepared by well-developed arc melting and casting method. Their microstructure and mechanical properties were examined, furthermore, we described an initial attempt to investigate the

* Corresponding author. Tel.: +86 010 62322160; fax: +86 010 62332508.
E-mail address: drzhangy@skl.ustb.edu.cn (Y. Zhang).

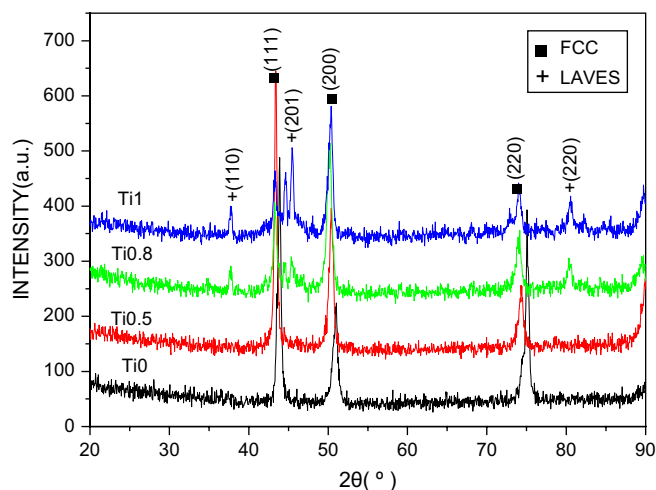


Fig. 1. XRD patterns of the CoCrCuFeNiTi_x rod samples with diameter of 5 mm ($x = 0, 0.5, 0.8$, and 1).

magnetic property of the alloy series above and discovered an interesting transition.

2. Experimental procedures

Ingots with nominal compositions of CoCrCuFeNiTi_x (x values in molar ratio, $x = 0, 0.5, 0.8$ and 1.0) were prepared from commercial-purity Ti (99.9%), Cr (99.9%), Fe (99%), Co (99.95%), Ni (99.9%) and Cu (99.9%) by arc melting under argon atmosphere on a water-cooled Cu hearth. The alloys

were remelted several times in order to improve homogeneity. Finally 5 mm diameter cylindrical rods were prepared in suction casting facility attached to the arc melter. Structural investigations were carried out by X-ray diffraction (XRD) with Cu K_α radiation using an MXP21VAHF diffractometer. An LEO-1450 scanning electron microscope (SEM) equipped with a KEVEX Sigma energy-dispersive X-ray (EDX) spectrometer was used for microstructure observation and phase composition identification. Thin slices from the as-cast rods were prepared by the conventional method of slicing and grinding, followed by dimpling and finally ion milling (LBS-1). These slices were used for detailed structural investigations of the as-cast samples by transmission electron microscopy (TEM). A CMT4305 testing machine was used for compressive tests, with the strain rate of 10^{-4} s^{-1} . In order to evaluate the magnetic property, squares of 3 mm length and 0.5 mm thickness were prepared from the 5 mm diameter as-cast rods and tested by an LDJ 9600 vibrating sample magnetometer (VSM). And the magnetic moments of low-temperature were characterized with an MPMS-7 superconducting quantum device magnetometer (SQUID Magnetometer).

3. Results

3.1. Microstructure evolution

The XRD patterns of the as-cast rods are presented in Fig. 1. Both CoCrCuFeNi and CoCrCuFeNiTi_{0.5} alloys show

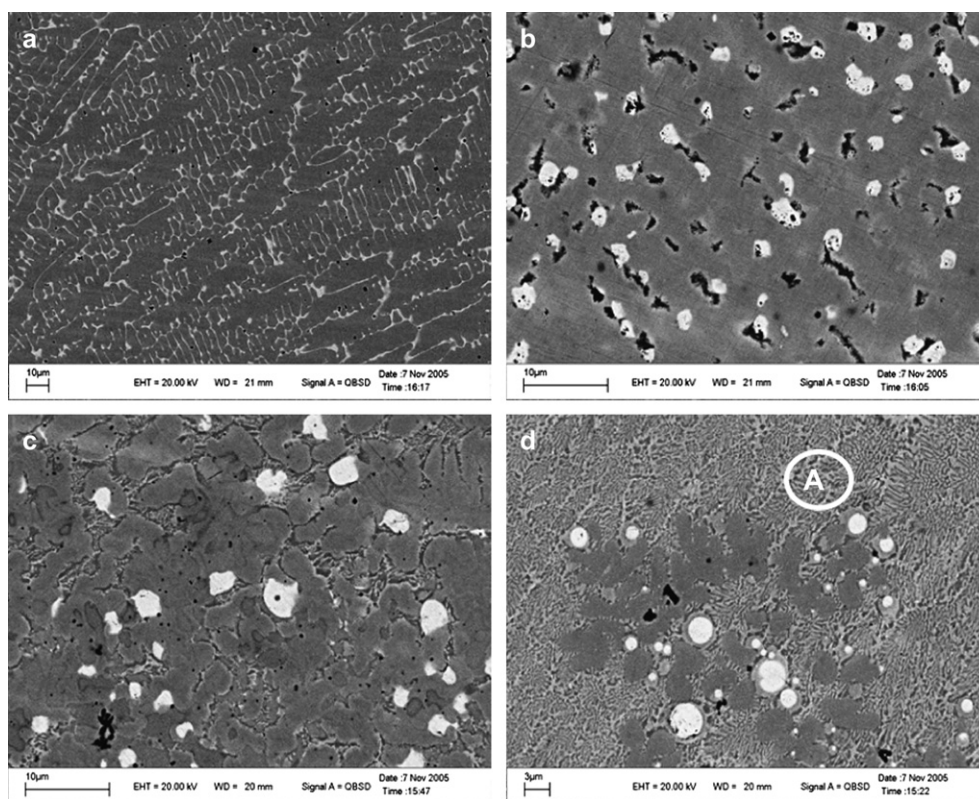


Fig. 2. SEM back-scatter electron images of the CoCrCuFeNiTi_x alloys after polishing, $x = 0, 0.5, 0.8$, and 1 corresponding to (a), (b), (c) and (d), respectively. In (d), a region marked by circle denotes the region for HRTEM observation (as shown in Fig. 3).

the presence of FCC reflections. For CoCrCuFeNiTi_{0.8} and CoCrCuFeNiTi, the reflections of Laves phase occur besides FCC reflections, the relative intensity of which is smaller than that of the FCC reflections. The lattice parameter of the solid solution is estimated to be 3.579 Å from the strongest (111) peak for CoCrCuFeNi. The energy-dispersive X-ray (EDX) analysis reveals that Cobalt acts as the solvent and other atoms replace the Cobalt atoms in FCC lattice. The Laves phase in the alloys of CoCrCuFeNiTi_{0.8} and CoCrCuFeNiTi, with lattice parameters of $a = 4.765$ Å and $c = 7.723$ Å, should be ranged as an Fe₂Ti type. The peaks of the FCC reflections shift a little leftwards with the increase of Ti addition, which indicates that the lattice parameter of FCC phase has a slight increment. A broad diffraction peak at $2\theta = 42\text{--}47^\circ$ for CoCrCuFeNiTi_{0.8} and CoCrCuFeNiTi alloys can be observed, which indicates the existence of amorphous structure.

The microstructure of the alloys is characterized in Fig. 2. The as-cast CoCrCuFeNi alloy exhibits a dendritic microstructure with an average primary arm width of about 15 μm and Cu element segregates preferentially to the interdendritic regions because more electronegative elements such as Cu are less stable in the FCC dendrites and are rejected into the interdendritic regions (as shown in Fig. 2(a)). Compared with CoCrCuFeNi alloy, the microstructure of CoCrCuFeNiTi_{0.5} alloy (as shown in Fig. 2(b)) displays a changed morphology. Both dendritic and interdendritic phases gathered to clusters of about 2–3 μm. According to EDX analysis of the black phase, it is reasonable to consider it as another FCC solid solution, but the solvent is Ni. In Fig. 2(c) and (d), Laves phase and eutectic can be observed, the difference lies in that eutectic only exists around crystal boundaries for CoCrCuFeNiTi_{0.8} alloy, while in CoCrCuFeNiTi alloy the eutectic structure has a rather high volume fraction.

Fig. 3 shows the HRTEM image of the A region marked by circle in Fig. 2(d). The disordered matrix, its inverse fast Fourier transform (IFFT) image (as shown in Fig. 3(c)), confirms the existence of amorphous phase. In addition, it can be seen that the dispersing nanoparticles, which may be NiTi₂ with an FCC lattice of $a = 3.553$ Å, embedded in the amorphous phase.

3.2. Mechanical properties

Fig. 4 shows the true stress–true strain curves of compression test for the alloy series. It is noticeable that CoCrCuFeNi alloy exhibits exceptional ductility so that the sample did not break even under about 50% true strain. And the alloy of CoCrCuFeNiTi_{0.5} possesses better integrated properties. The maximum compressive strength of CoCrCuFeNiTi_{0.5} alloy is 1650 MPa. It exhibits about 22.4% deformation and a large work hardening capacity in the compression test at room temperature. Compared with CoCrCuFeNi, the CoCrCuFeNiTi_{0.5} alloy has an enhanced strength, but lower ductility, which can be ascribed to the reason that the second-phase particles staggering with matrix lattice and it requires extra energy for the

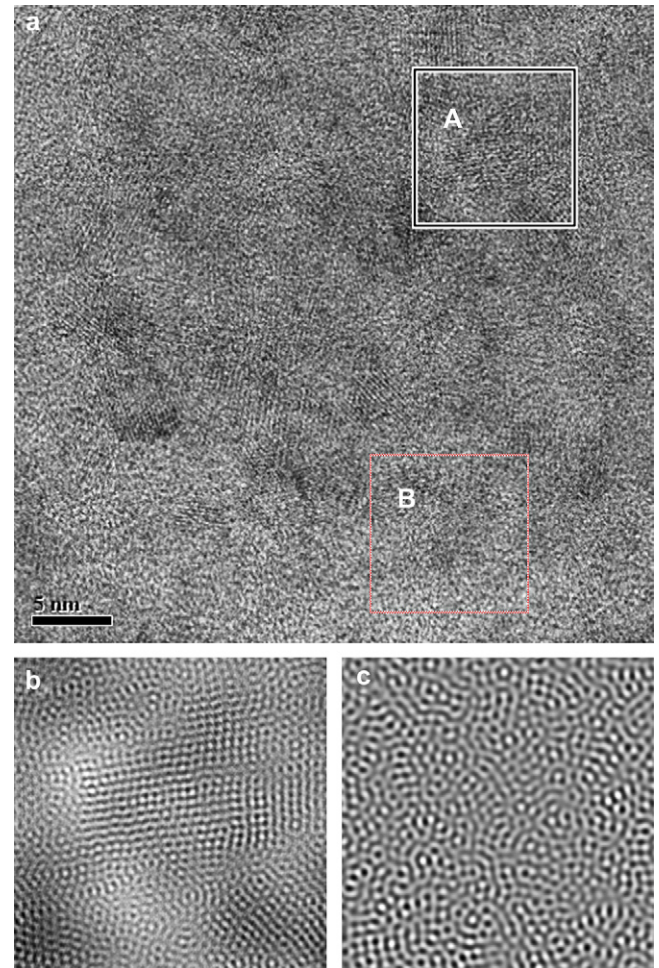


Fig. 3. (a) HRTEM image of nanoparticles embedded in the amorphous phase in CoCrCuFeNiTi alloy, its IFFT images (b) and (c) indicate, respectively, region A of nanoparticles and region B of full amorphous matrix. The inset shows the diffraction pattern of the nanophase.

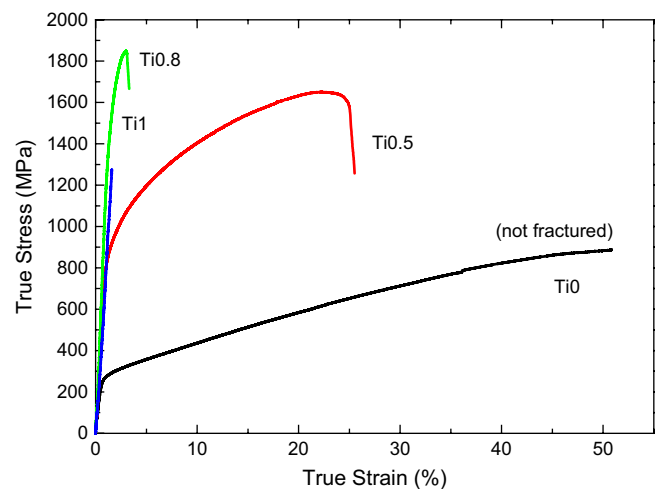


Fig. 4. True stress–true strain curves of the CoCrCuFeNiTi_x rod samples with diameter of 5 mm ($x = 0, 0.5, 0.8$, and 1).

Table 1
Room temperature compression test results for the alloy series

Alloy	E (GPa)	σ_y (MPa)	$\sigma_{\max}$ (MPa)	ε_e (%)	ε_p (%)
CoCrCuFeNi	55.6	230	888	0.56	50.2
CoCrCuFeNiTi _{0.5}	98.6	700	1650	0.8	21.6
CoCrCuFeNiTi _{0.8}	128.3	1042	1848	0.89	2.11
CoCrCuFeNiTi	76.5	1272	1272	1.6	0

Young's modulus, E ; yield stress, σ_y ; compressive strength, $\sigma_{\max}$; elastic strain limit, ε_e ; and plastic strain limit, ε_p .

dislocations passing through the second-phase particles. With Ti content increasing to $x = 0.8$, the maximum compressive strength has a slight increase to 1848 MPa, but the deformation sharply decreases to 3%. As for CoCrCuFeNiTi alloy, it has only elastic strain of 1.6% and its fracture strength is 1272 MPa. The Young's modulus (E), yield strength σ_y , compressive strength $\sigma_{\max}$, elastic strain limit ε_e and plastic strain limit ε_p are listed in Table 1.

The specimen surface and the fracture surfaces after failure are presented in Fig. 5. Fig. 5(a) shows the side surface of the as-cast CoCrCuFeNi cylinder sample. On the surface there are a large number of slip bands which are inclined to horizontal direction. For CoCrCuFeNiTi_{0.5} alloy, we can see clusters of dendrites along various directions in Fig. 5(b), which might be attributed to its favorable ductility. Fig. 5(c) shows the fracture surface of CoCrCuFeNiTi_{0.8} alloy. Due to the existence of brittle microstructure around the crystal boundaries, the plasticity of the alloy becomes very weak. It can be seen that the fracture pattern of CoCrCuFeNiTi alloy is similar to that of CoCrCuFeNiTi_{0.8} alloy in Fig. 5(d), and the brittle structure accounts for a significant proportion, which results in CoCrCuFeNiTi alloy losing plasticity at all.

3.3. Magnetic properties

Fig. 6 presents the magnetization curves of the alloy series. Both CoCrCuFeNi and CoCrCuFeNiTi_{0.5} exhibit typical paramagnetism, and the saturation magnetizations are 1.505 emu/g and 0.333 emu/g, respectively, under 20,000 Oe magnetic field, while the alloys of CoCrCuFeNiTi_{0.8} and CoCrCuFeNiTi have likely superparamagnetic curves with the saturation magnetizations of 1.368 emu/g and 1.511 emu/g, respectively. For the high resolution transmission electron microscopy (HRTEM) observation (as shown in Fig. 3), amorphous phase and crystalline phase will exhibit different phasic contrasts. According to the scale given, the sizes of the nanoparticles are around 3–4 nm, which ensure granule volumes small enough to exhibit superparamagnetic property [13].

4. Discussion

According to Greer's 'confusion principle': the more elements involved, the lower the chance that the alloy can select viable crystal structures, and the greater the chance of glass formation [12]. And the further explanation is that the most 'confusing' elements will be those that differ most from each other in size because a mixture of these elements lead to dense packing in the liquid state, and the resulting stability favours glass formation over crystallization. The little difference in size is just the reason why the 'confusion principle' is not applicable in our alloy series when x is less than 0.5. On the other hand, the formation of solid solution still follows the principle of lowest energy. According to the formula of free energy: $G = H - TS$, solid solution containing multiprincipal elements will tend to be more stable at elevated temperatures because

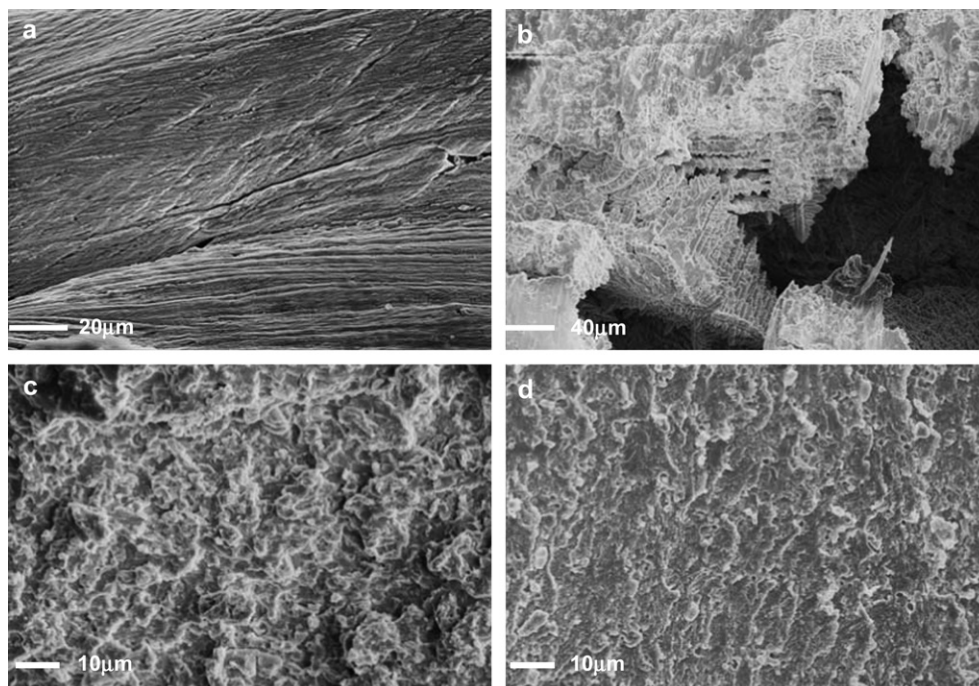


Fig. 5. SEM secondary electron images of fracture surfaces of the CoCrCuFeNiTi_x alloys, $x = 0, 0.5, 0.8$, and 1 corresponding to (a), (b), (c) and (d), respectively.

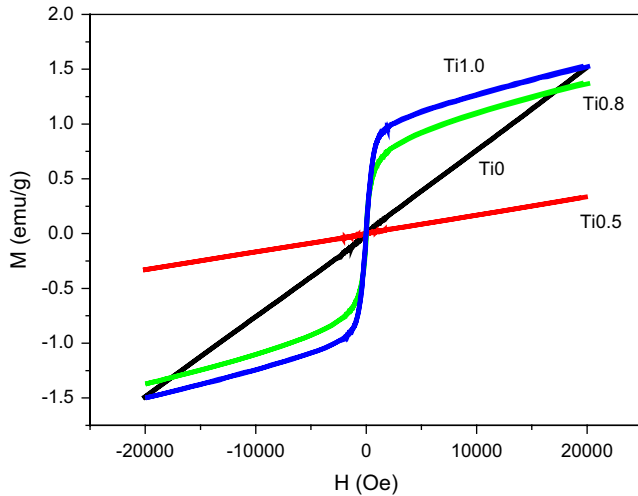


Fig. 6. Magnetization curves of the $\text{Ti}_x\text{CoCrCuFeNi}$ alloys ($x = 0, 0.5, 0.8$, and 1), showing a transition from the paramagnetism ($x = 0$ and 0.5) to superparamagnetism ($x = 0.8$ and 1).

of the large mixing entropies [10]. To check the formula of mixing entropy: $S_m = -R \sum_{i=1}^n C_i \ln C_i$, a maximum point can be attained when the alloy system contains equiatomic multicomponent. With the increase of Ti addition, a dense packing will be attained in local regions due to the proper arrangement of elements with remarkable difference in atomic size such as Ti and Ni, the difference in atomic radii is near 17%. Hence, the ‘confusion principle’ may be applicable in this case and glass formation does occur in local regions.

It can be seen that both the onset melting temperature T_m and the liquidus temperature T_l decrease with increasing Ti addition (Fig. 7). This tendency should be derived from the evolution of the microstructure. Based on the fact that almost all of the binary phase diagrams of Ti–M (M refer to Co, Cu, Fe and Ni) are eutectic type, and their eutectic temperatures are very close to the T_m of $\text{CoCrCuFeNiTi}_{0.8}$ and CoCrCuFeNiTi alloys, so it may be reasonable to conceive a pseudo-binary eutectic diagram (Fig. 8) for illustrating the formation of

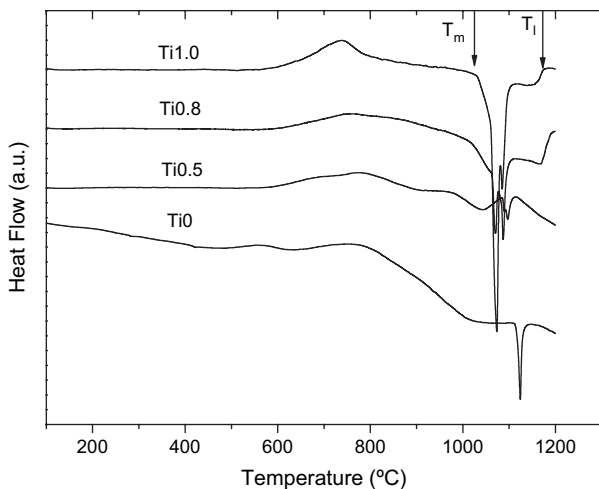


Fig. 7. DSC heating curves of the CoCrCuFeNiTi_x alloys ($x = 0, 0.5, 0.8$ and 1.0).

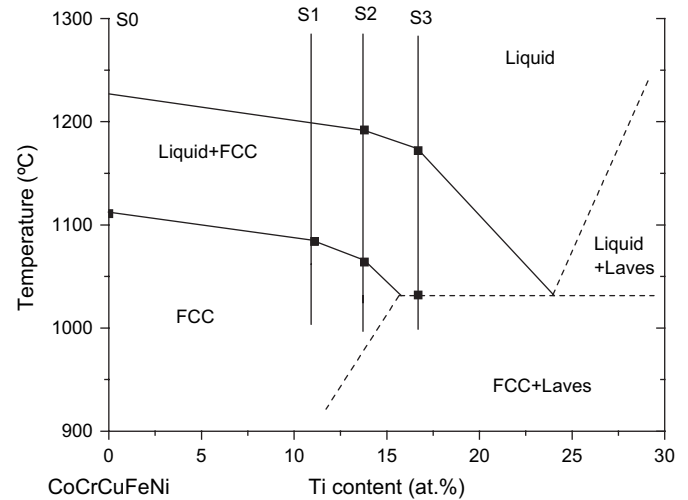


Fig. 8. The diagrammatic sketch of CoCrCuFeNi system with the Ti addition, S0, S1, S2 and S3 corresponding to $x = 0, 0.5, 0.8$ and 1.0 , respectively.

microstructure. According to Fig. 8, the systems of $\text{CoCrCuFeNiTi}_{0.8}$ and CoCrCuFeNiTi alloys (marked S2 and S3 in Fig. 8) would precipitate FCC solid solution as primary phase, then eutectic resultant including Laves phase of Fe_2Ti type because of the non-equilibrium solidification resulting from high cooling rate. And the local segments having the composition close to eutectic may favour a glass transition under high cooling rates. As a result of fast cooling, the short-range structure (denoted by arrows in Fig. 3(c)) of the amorphous phase showing a hexagonal pattern which accords with the lattice of Fe_2Ti type.

In general, a number of slip bands would be beneficial for plasticity. For CoCrCuFeNi and $\text{CoCrCuFeNiTi}_{0.5}$, the dislocations cannot propagate through the whole sample due to the constraint of the interdendritic regions. And multiple slip bands may be activated resulting from the stress concentration in this case, thus the ductility improved. Both Fig. 5(c) and (d) show the typical fracture surfaces of quasi-cleavage facets, and also a few dimples (Fig. 5(d)). The difference lies in the facets in Fig. 5(d) which are bigger and flatter reflecting their weaker ductility.

The fact of magnetic transition can also be proved by the temperature dependence of magnetization ($M(T)$) curves presented in Fig. 9. For CoCrCuFeNi and $\text{CoCrCuFeNiTi}_{0.5}$, there are only Curie temperatures (T_C) appearing on the curves, whereas the alloys of $\text{CoCrCuFeNiTi}_{0.8}$ and CoCrCuFeNiTi have the blocking temperatures (T_B) besides T_C in a lower temperature region of 19–23 K. Despite T_B is so low that it limits the application for this kind of superparamagnetic alloys, the tendency of T_B increasing with increased addition of Ti is irradiative for further improving the magnetic properties by adjusting the alloy composition.

5. Conclusions

In summary, the alloy series of CoCrCuFeNiTi_x (x values in molar ratio, $x = 0, 0.5, 0.8$ and 1.0) have relatively simple

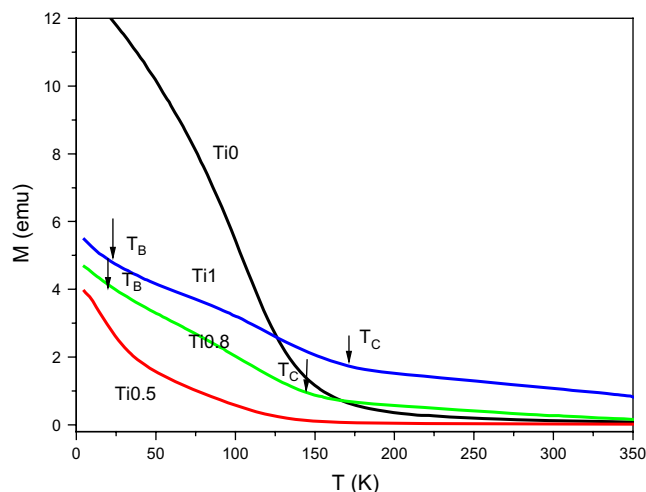


Fig. 9. The temperature dependence of magnetization ($M(T)$) curves while cooling the alloy series of CoCrCuFeNiTi_x ($x = 0, 0.5, 0.8$ and 1) at 2000 Oe .

microstructure and promising properties. As Ti addition increases from 0 to 1.0, the microstructure of the alloy series evolves from a single FCC solid solution to the composite structure: FCC and Laves phases, also a fraction of amorphous phase. The yield strength of the alloys increases from 230 MPa to 1272 MPa with the increase of Ti addition, among which the $\text{CoCrCuFeNiTi}_{0.5}$ alloy exhibits high strength of up to 1650 MPa together with extensive work hardening and large ductility of 22%. For CoCrCuFeNi and $\text{CoCrCuFeNiTi}_{0.5}$ alloys, the FCC solid solutions exhibit typical paramagnetism, while $\text{CoCrCuFeNiTi}_{0.8}$ and CoCrCuFeNiTi alloys exhibit superparamagnetism which may be attributed to the

appearance of nanoparticle assemblies embedded in the amorphous phase due to the increment of Ti addition. And the increase of T_B with Ti addition may suggest a way for developing a new superparamagnetic material.

Acknowledgment

The authors would like to thank X.P. Song for technical assistance in TEM experiments, and X.J. Liu and J. Chen for processing the figures. One of the authors (Z.Y.) would like to acknowledge the financial support by National Natural Science Foundation of China (NNSFC) under the granted No. 50571018, and Program for New Century Excellent Talents in University (NCET).

References

- [1] Cantor B, Chang ITH, Knight P, Vincent AJB. *Mater Sci Eng A* 2004;375-377:213.
- [2] Porter DA, Easterling KE. *Phase transformations in metals and alloys*. 2nd ed. Nelson Thornes Ltd.; 2001.
- [3] Liu CT, Stiegler JO. *Science* 1984;226(4675):636.
- [4] Chen GL, Liu CT. *Int Mater Rev* 2001;46(60):253.
- [5] Inoue A. *Mater Sci Eng A* 2004;375-377:16.
- [6] Wang WH, Dong C, Shek CH. *Mater Sci Eng R* 2004;44(2-3):45.
- [7] Tan H, Zhang Y, Ma D, Feng YP, Li Y. *Acta Mater* 2003;51:4551.
- [8] Johnson WL. *MRS Bull* 1999;24:42.
- [9] Zhang Y, Zhao DQ, Pan MX, Wang WH. *J Non-Cryst Sol* 2003;315:206.
- [10] Tong CJ, Chen YL, Chen SK. *Met Mater Trans A* 2005;36A:881.
- [11] Tong CJ, Chen MR, Chen SK. *Met Mater Trans A* 2005;36A:1263.
- [12] Greer AL. *Nature* 1993;366:303.
- [13] Shinde SR, Ogale SB, Higgins JS. *Phys Rev Lett* 2004;92:166601.